Ubuntu-Connect

AI-powered early warning for students at risk of failing

VAAL UNIVERSITY OF TECHNOLOGY
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Business Analysis 3.2 (AIBUY3A) | Vaal University of Technology | Theme: An AI Solution for Industries (Education)

The Problem & Our Solution

The Problem:

Staff usually learn that a student is failing only after end-of-term results, when little time is left to help. Marks, attendance and assignment records already exist but are rarely analyzed together, so early warning signs go unnoticed. The result is avoidable failure, dropout, and wasted fees.

Our Solution:

Ubuntu-Connect scores each student's risk from data the institution already holds and explains every Low, Medium or High flag in plain language.

Tech Stack:

Python machine learning, grade-trend features, teacher note sentiment (NLPI), and a FastAPI service with a rule-based chatbot (all open source).

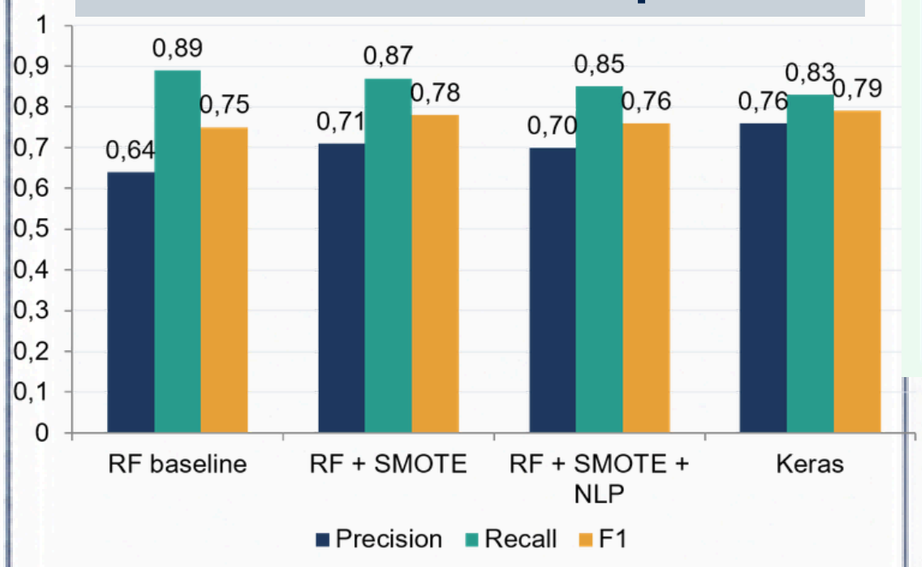
5-step process

1. **DATA** — 1,044 students, 33 attributes
2. **FEATURES** — G1 to G2 trend, average, sentiment
3. **MODELS** — Random Forest, SMOTE, Keras
4. **RISK API** — High/Medium/Low + reason
5. **CHATBOT** — Tips for staff via /chat

|Data: Cortez and Silva (2008), UCI Student Performance dataset

Key Results & Metrics

Model Perfomance Comparison



Key Results

- High-risk flags: precision 0.76 (35 of 46), above the 70% target.
- Medium and High flags together catch 93% of at-risk students.
- Deployed model: Random Forest + SMOTE (no NLP) with precision 0.71 and recall 0.87.
- Teacher note sentiment (NLP) did not help: F1 dropped from 0.78 to 0.76.
- Every Medium and High flag carries a plain-language reason (checked on all 1,044 students).

Pipeline Progress

Pipeline Progress

- [x] Person 1: Load & combine data — merged to main
- [x] Person 2: Feature Engineering (feature_engineering.py) #1 Person 2: Feature Eng
- [x] Person 3: Random Forest baseline #2 Person 3: Random Forest baseline
- [x] Person 4: Deep Learning model #3 Person 4: Deep Learning model
- [x] Person 5: NLP Sentiment #4 Person 5: NLP Sentiment
- [x] Person 6: SMOTE + Final Model #5 Person 6: SMOTE + Final Model
- [x] Person 7: API /predict endpoint #6 Person 7: API /predict endpoint
- [x] Person 8: API /chat endpoint #7 Person 8: API /chat endpoint
- [x] Person 9: Documentation & Poster #8 Person 9: Documentation & Poster

API Demo, Limits, & Team

/predict Endpoint

Response body


```
{
  "student_id": "73",
  "at_risk": true,
  "risk_probability": 1,
  "risk_level": "High",
  "reason": "Main factors: average of G1 and G2
rior failures = 2 (above average); change in gra
average)"
}
```


Input : {
"student_id": "73","features": {"age": 15.0,"studytime": 2.0,"failures": 2.0,"absences": 2.0,"g1_to_g2_change": -2.0,"avg_g1_g2": 7.0,"address": 0.0,"famsize": 0.0, "Pstatus": 1.0,"schoolsup": 1.0,"famsup": 1.0 }}

/chat Endpoint


```
{
  "reply": "Student 73: High risk (probability 1.00).\nSuggested next step:
ontact the student this week, and record the intervention so its effect can
e measured.\nTips:\n1. Arrange extra support for the weaker topics (tutorin
past papers, office hours) and set one small target for the next assessmen
t.\n2. Review the earlier failed subject(s) with the student, find the topi
that caused the problem, and offer catch-up tutoring or a peer study partne
r.\n3. Grades are falling between periods: ask what changed and review the
st recent assessments to see where marks dropped.\nThese are suggestions fo
staff. Confirm with the student before acting: the risk score is a prompt t
talk, not a verdict.",
  "risk_level": "High",
  "risk_probability": 1,
  "tips": [
    "Arrange extra support for the weaker topics (tutoring, past papers, of
ce hours) and set one small target for the next assessment.",
    "Review the earlier failed subject(s) with the student, find the topics
hat caused the problem, and offer catch-up tutoring or a peer study partne
r.",
    "Grades are falling between periods: ask what changed and review the mo
recent assessments to see where marks dropped."
  ]
```


> *These are suggestions for staff. Confirm with the student before acting: the risk score is a prompt to talk, not a verdict."*

Limits & Next Steps

Only 46 at-risk students in the test set: teacher notes are "synthetic". Age, address, family size and parents' status need a fairer review. Next: Pilot with consented data, staff dashboard, measure fail-rate reduction.

Team

Sphamandla Radebe (Data loading), Siyabonga Nzotho (Feature engineering), Pontsho Radebe (RF baseline), Prudence Mbele (Deep Learning/Keras), Khutso Boloka (NLP), Humphrey Maluleke (SMOTE/Final model), Esnath Kwindu (API predict), Khanimamba Maluleke (Chatbot/chat), Bayede Mthethwa (Documentation/poster).